

Selected Tree Queries and LDBC SNB SF1

Graph	Query	Parameters	Apache AGE					Kuzu					Neo4j				
			B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P	B (ms)	D	P	S _D	S _P
F1K	<i>Q_{desc}</i>	Root	441.6	2.3	2.3	190x	190x	6.3	3.1	2.5	2.1x	2.5x	29.5	28.5	41.5	1.0x	0.71x
F1K	<i>Q_{desc}</i>	Leaf parent	13.0	1.9	1.8	6.7x	7.1x	3.6	2.0	2.0	1.7x	1.8x	1.8	1.7	1.9	1.0x	0.95x
WT100	<i>Q_{desc}</i>	Root	26.0	2.2	1.8	12x	14x	3.5	1.5	1.5	2.3x	2.3x	11.6	13.2	13.5	0.88x	0.86x
WT100	<i>Q_{desc}</i>	Leaf parent	11.7	1.7	1.7	6.7x	6.8x	3.0	1.5	1.5	2.0x	2.0x	1.7	2.1	1.8	0.82x	0.95x
WT1K	<i>Q_{desc}</i>	Root	1,152.5	2.6	2.7	440x	420x	8.7	1.6	2.2	5.3x	3.9x	58.8	45.1	46.8	1.3x	1.3x
WT1K	<i>Q_{desc}</i>	Leaf parent	21.9	2.3	1.7	9.7x	13x	3.5	1.6	1.7	2.2x	2.0x	2.2	2.3	2.5	0.93x	0.88x
WT10K	<i>Q_{desc}</i>	Root	112,250.6	8.6	10.9	13Kx	10Kx	24.9	10.3	8.6	2.4x	2.9x	407.5	486.5	503.7	0.84x	0.81x
WT10K	<i>Q_{desc}</i>	Leaf parent	114.7	5.7	1.9	20x	60x	3.0	4.6	2.9	0.65x	1.0x	2.4	2.6	2.6	0.92x	0.93x
WT100K	<i>Q_{desc}</i>	Root	23,750,824.0	76.4	74.2	310Kx	320Kx	122.5	91.6	75.6	1.3x	1.6x	4,313.0	5,124.1	5,269.5	0.84x	0.82x
WT100K	<i>Q_{desc}</i>	Leaf parent	575.1	37.4	1.9	15x	300x	2.2	50.7	26.9	0.04x	0.08x	1.2	1.3	1.3	0.93x	0.92x
SNB/C	<i>Q_{desc}</i>	Root	91,997.5	681.0	2.0	140x	47Kx	3.6	417.0	391.4	0.01x	0.01x	4.7	4.5	4.7	1.0x	1.0x
SNB/C	<i>Q_{desc}</i>	Leaf parent	22,391.0	724.1	2.0	31x	11Kx	3.4	436.1	392.9	0.01x	0.01x	1.6	1.6	1.6	1.0x	0.99x
SNB/P	<i>Q_{desc}</i>	Root	16,838.8	2.7	2.6	6.2Kx	6.6Kx	6.9	2.5	2.3	2.8x	3.1x	46.3	41.2	42.9	1.1x	1.1x
SNB/P	<i>Q_{desc}</i>	Leaf parent	14,506.6	1.6	1.9	9.0Kx	7.8Kx	3.8	2.1	1.9	1.8x	2.0x	1.1	1.3	1.5	0.87x	0.74x
SNB/T	<i>Q_{desc}</i>	Root	14,465.3	1.9	1.9	7.7Kx	7.6Kx	3.3	1.7	1.6	2.0x	2.1x	9.7	11.0	11.3	0.89x	0.85x
SNB/T	<i>Q_{desc}</i>	Leaf parent	14,497.3	1.6	1.7	9.0Kx	8.6Kx	2.9	1.6	1.5	1.9x	2.0x	1.5	1.5	1.5	1.0x	1.0x
F1K	<i>Q_{a&d}</i>	Root-leaf	12.5	1.4	1.5	8.8x	8.4x	19.1	1.0	0.9	19x	21x	1.7	1.1	1.6	1.5x	1.1x
F1K	<i>Q_{a&d}</i>	Deep parent-leaf	12.2	1.6	1.5	7.5x	8.4x	21.6	1.0	0.9	22x	24x	1.8	1.2	1.6	1.6x	1.2x
WT100	<i>Q_{a&d}</i>	Root-leaf	11.4	1.7	1.6	6.7x	7.1x	19.1	0.9	0.9	21x	20x	1.4	1.0	1.1	1.4x	1.3x
WT100	<i>Q_{a&d}</i>	Deep parent-leaf	11.1	1.5	1.5	7.5x	7.6x	18.3	0.9	0.9	20x	19x	1.2	1.1	1.3	1.1x	0.98x
WT1K	<i>Q_{a&d}</i>	Root-leaf	12.7	1.6	1.5	8.1x	8.3x	20.7	0.8	0.9	25x	23x	1.1	0.8	1.1	1.3x	0.94x
WT1K	<i>Q_{a&d}</i>	Deep parent-leaf	12.1	1.5	1.5	7.8x	8.0x	18.6	0.8	0.9	22x	21x	1.3	1.3	1.3	1.0x	0.99x
WT10K	<i>Q_{a&d}</i>	Root-leaf	19.1	1.7	1.8	11x	11x	21.2	0.9	0.9	25x	25x	1.4	1.2	0.9	1.2x	1.5x
WT10K	<i>Q_{a&d}</i>	Deep parent-leaf	18.5	1.8	1.7	11x	11x	21.0	1.0	0.9	22x	23x	1.2	1.1	1.1	1.1x	1.1x
WT100K	<i>Q_{a&d}</i>	Root-leaf	79.8	1.9	1.8	42x	44x	20.2	0.9	0.9	22x	23x	1.6	1.3	1.3	1.2x	1.2x
WT100K	<i>Q_{a&d}</i>	Deep parent-leaf	81.2	1.8	1.8	44x	45x	22.5	0.8	0.9	27x	25x	1.3	1.3	1.3	1.0x	0.96x
SNB/C	<i>Q_{a&d}</i>	Root-leaf	14,648.6	2.0	1.9	7.3Kx	7.9Kx	22.3	1.0	0.9	22x	24x	1.6	1.2	1.1	1.3x	1.4x
SNB/C	<i>Q_{a&d}</i>	Deep parent-leaf	14,582.5	2.1	1.8	6.9Kx	8.0Kx	19.4	1.3	1.0	15x	19x	1.2	1.1	1.1	1.1x	1.0x
SNB/P	<i>Q_{a&d}</i>	Root-leaf	14,525.6	1.1	1.5	13Kx	9.7Kx	19.6	0.8	0.9	24x	22x	1.4	1.1	1.1	1.3x	1.3x
SNB/P	<i>Q_{a&d}</i>	Deep parent-leaf	14,422.9	1.6	1.5	9.2Kx	9.7Kx	17.3	0.9	0.9	20x	19x	1.3	1.1	1.1	1.2x	1.1x
SNB/T	<i>Q_{a&d}</i>	Root-leaf	14,538.5	1.6	1.7	9.0Kx	8.6Kx	18.9	0.8	0.9	23x	20x	1.2	1.1	1.2	1.1x	1.0x
SNB/T	<i>Q_{a&d}</i>	Deep parent-leaf	14,437.5	1.4	1.7	10Kx	8.6Kx	20.5	0.8	0.9	25x	23x	1.4	1.1	1.1	1.3x	1.3x
SNB	IC12	-	188,529.8	170,910.6	173,980.7	1.1x	1.1x	396.5	304.9	403.9	1.3x	0.98x	157.5	119.4	106.3	1.3x	1.5x
SNB	IS2	-	61,383.9	3,842.9	13,060.7	16x	4.7x	43.7	242.2	277.4	0.18x	0.16x	4.5	4.5	10.7	0.99x	0.42x
SNB	IS6	-	17,296.9	2,172.2	3,269.6	8.0x	5.3x	28.8	22.9	118.4	1.3x	0.24x	1.9	2.0	1.8	0.95x	1.0x

Queries: *Q_{desc}* = All Descendants; *Q_{a&d}* = Check if Ancestor; IC12 = Interactive Complex 12; IS2 = Interactive Short 2; IS6 = Interactive Short 6.

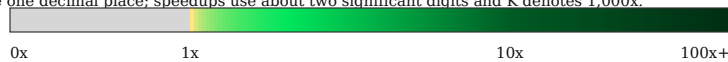
Parameters: Root = root of largest tree; Leaf parent = parent of a leaf; Root-leaf = root-farthest-leaf pair; Deep parent-leaf = deepest parent-leaf pair.

Methods: B = Baseline; D = Dewey; P = PrePost; S_D = B / D; S_P = B / P. All times are in ms.

Graphs: F = forest; WT = ultrawide tree; K = 1,000 nodes; SNB/C, SNB/P, SNB/T = SNB SF1 trees; SNB = full SNB SF1 graph.

Rounding: runtimes use one decimal place; speedups use about two significant digits and K denotes 1,000x.

Speedup color:



gray = slowdown; yellow = 1x; green = speedup; darker green = larger speedup